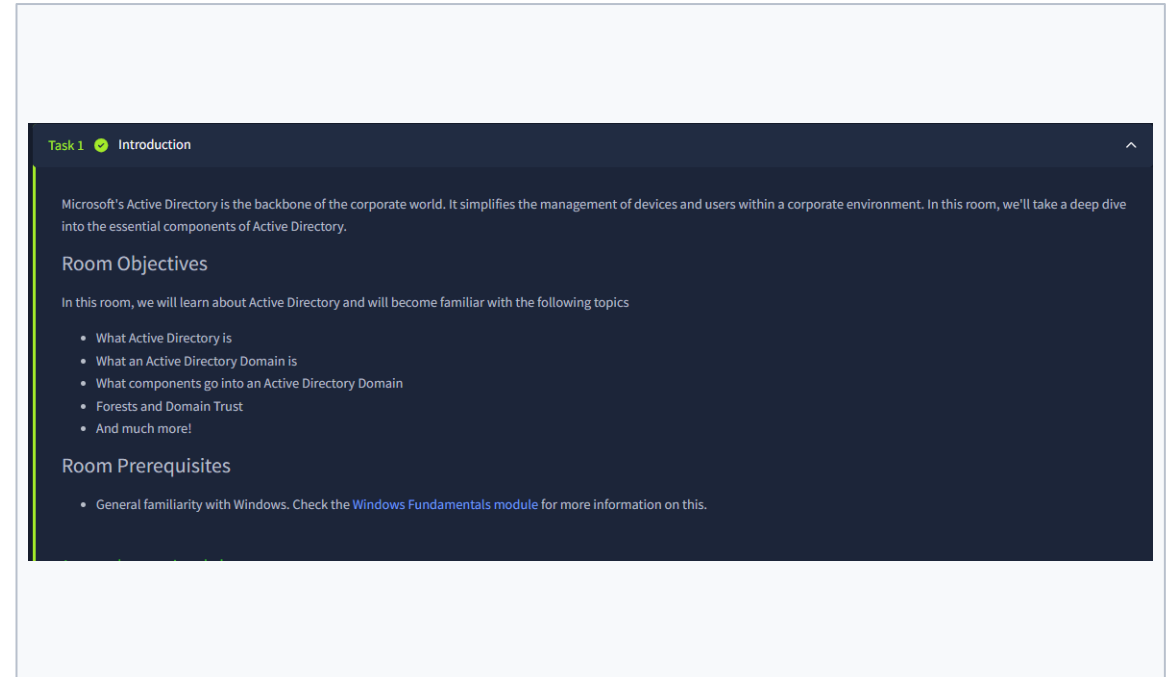


Active Directory Administration Home Lab

Guided TryHackMe virtual environment

Jerrell Hawkins

Windows Server | AD DS | Group Policy | PowerShell | RDP



Lab objective

Practice Active Directory domains, OU's and user administration, delegated access, Group Policy, and remote validation.

The Lab Simulated Core Tier 1 Domain Administration Work

Environment

Windows Server domain inside a guided TryHackMe virtual machine; administration performed through Microsoft management consoles, RDP, and PowerShell.

Access control

Delegated only the Sales password-reset task to Phillip instead of granting broad administrative membership.

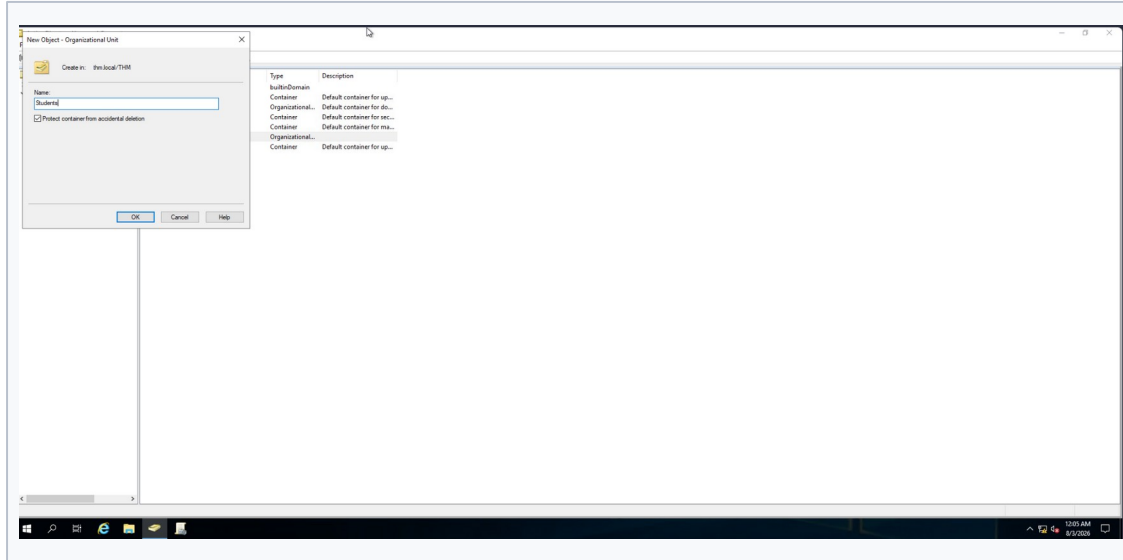
Directory operations

Created an OU, aligned users to department OUs, and removed an obsolete OU after intentionally disabling deletion protection.

Policy and validation

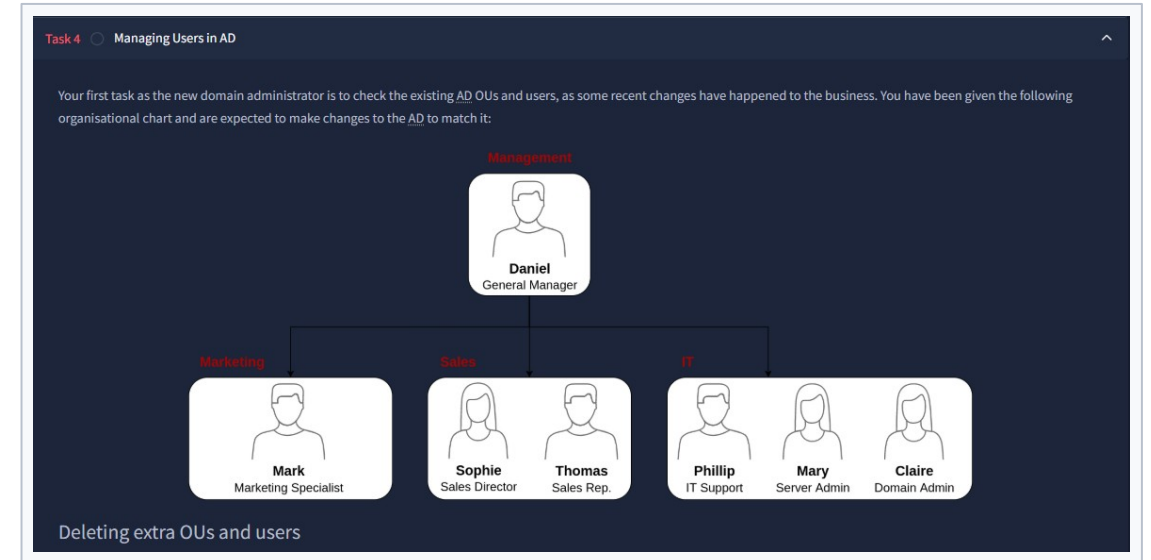
Configured password and workstation controls, linked GPOs at the correct scope, and tested the result from a standard user's session.

I Rebuilt the Directory Structure to Match the Business Model



Created the Students OU

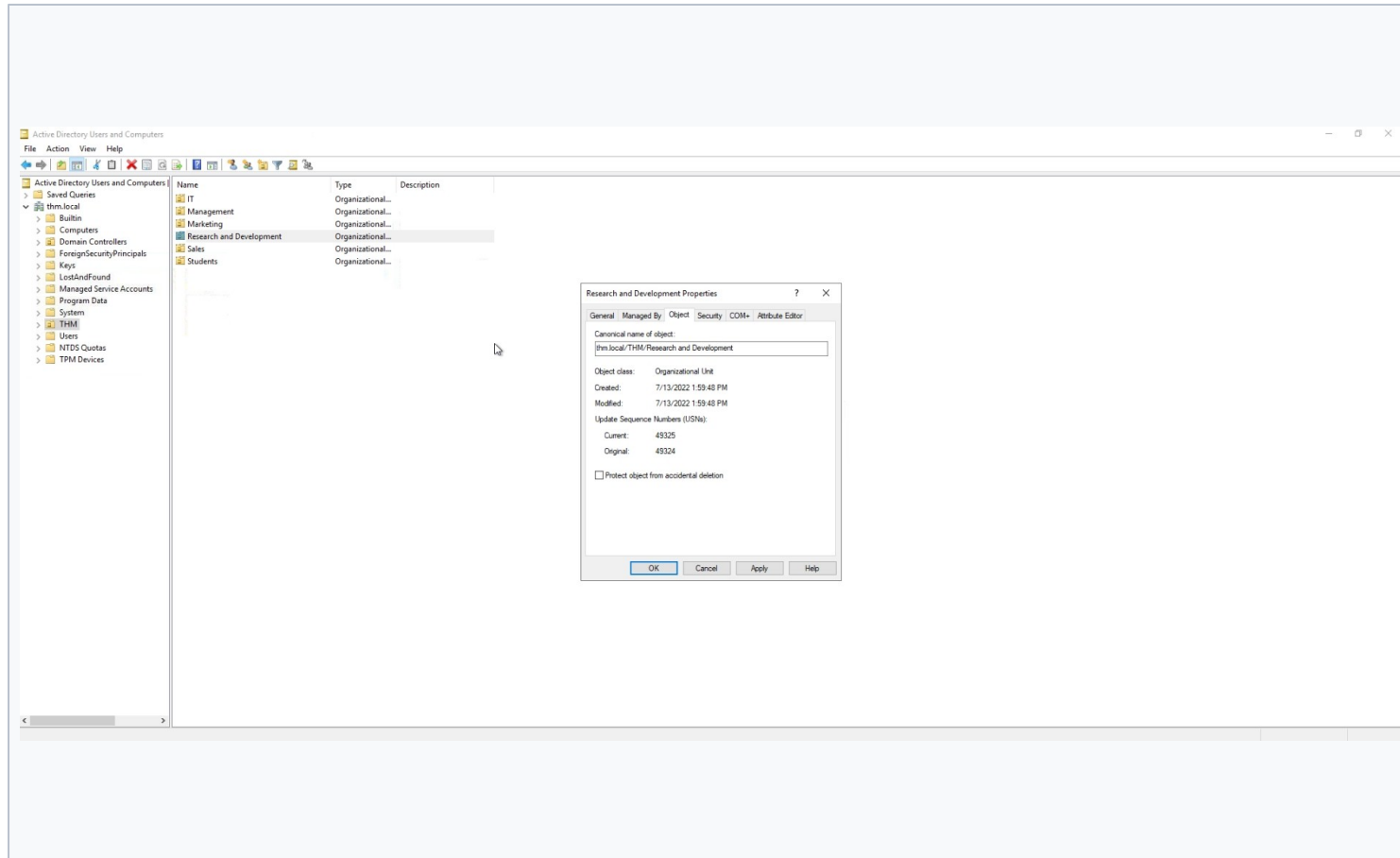
In Active Directory Users and Computers (ADUC), I right-clicked the domain container, selected New > Organizational Unit, entered Students, and kept accidental-deletion protection enabled.



Aligned objects to the target state

I used the organizational chart as the desired state, reviewed existing OUs and user accounts, and reorganized or removed objects so Management, Marketing, Sales, and IT matched their assigned roles.

I Removed an Obsolete OU only after Disabling its Safety Control



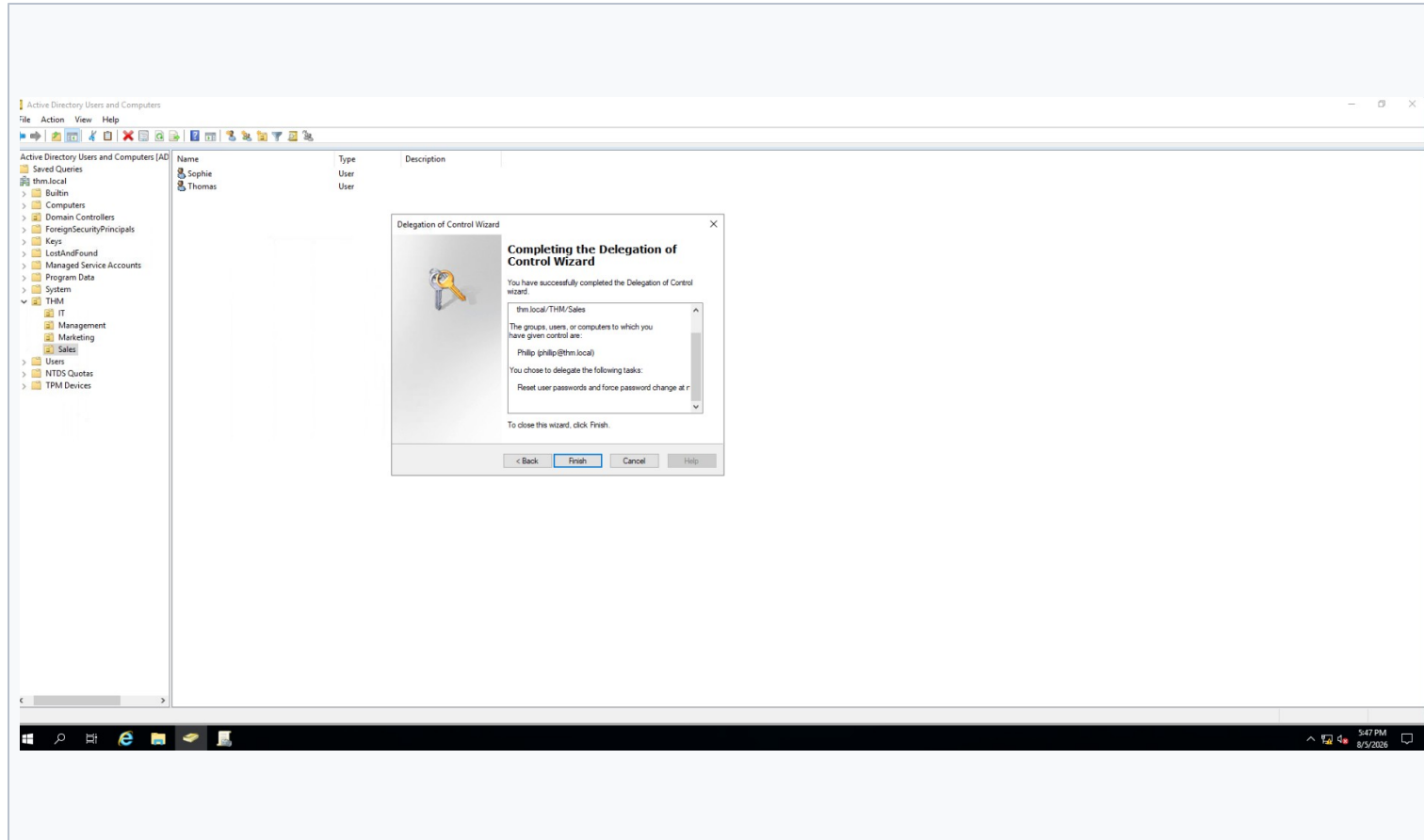
Resolution

- Enabled View > Advanced Features in ADUC.
- Opened the R&D OU's Properties and selected the Object tab.
- Cleared Protect object from accidental deletion.
- Deleted the obsolete OU because it was not part of the approved target structure.

Technical correction

This action removed a deletion-protection flag. It did not grant the administrator new privileges.

I Delegated Password Resets without Granting Broad Admin Rights



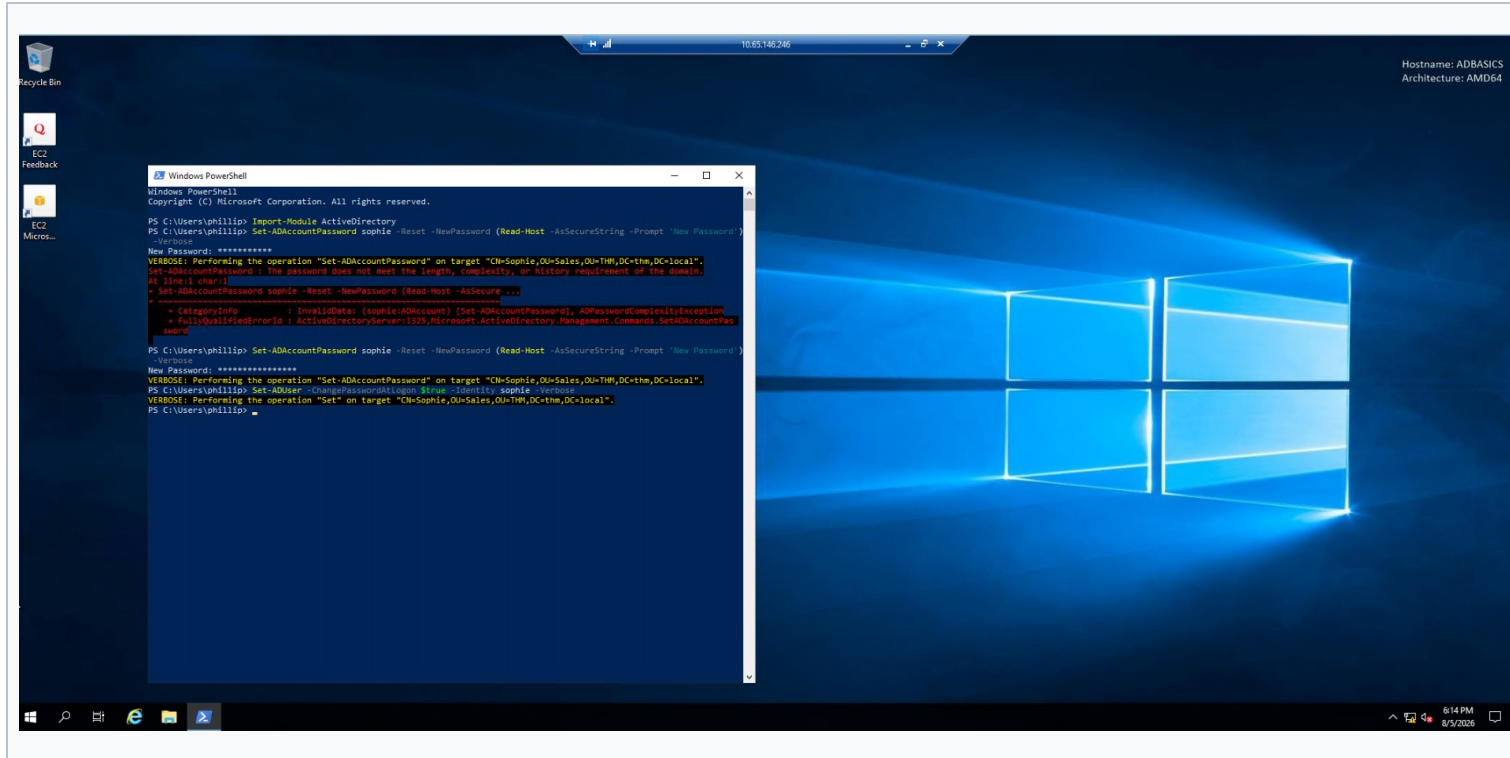
Delegation path

- Right-clicked the Sales OU and opened Delegate Control.
- Added THM\phillip and confirmed the account name.
- Selected Reset user passwords and force password change at next login.
- Completed the wizard and scoped the permission to the Sales OU.

Why it matters

Phillip received one support capability for one department - a practical example of least privilege.

I Corrected a Failed Delegated Password Reset



What the evidence shows

- Connected over RDP as THM\phillip and imported the ActiveDirectory module.
- The first password was rejected because it did not meet domain policy.
- Retried with another password; the reset succeeded.
- Used a separate Set-ADUser command to require a password change at Sophie's next logon.

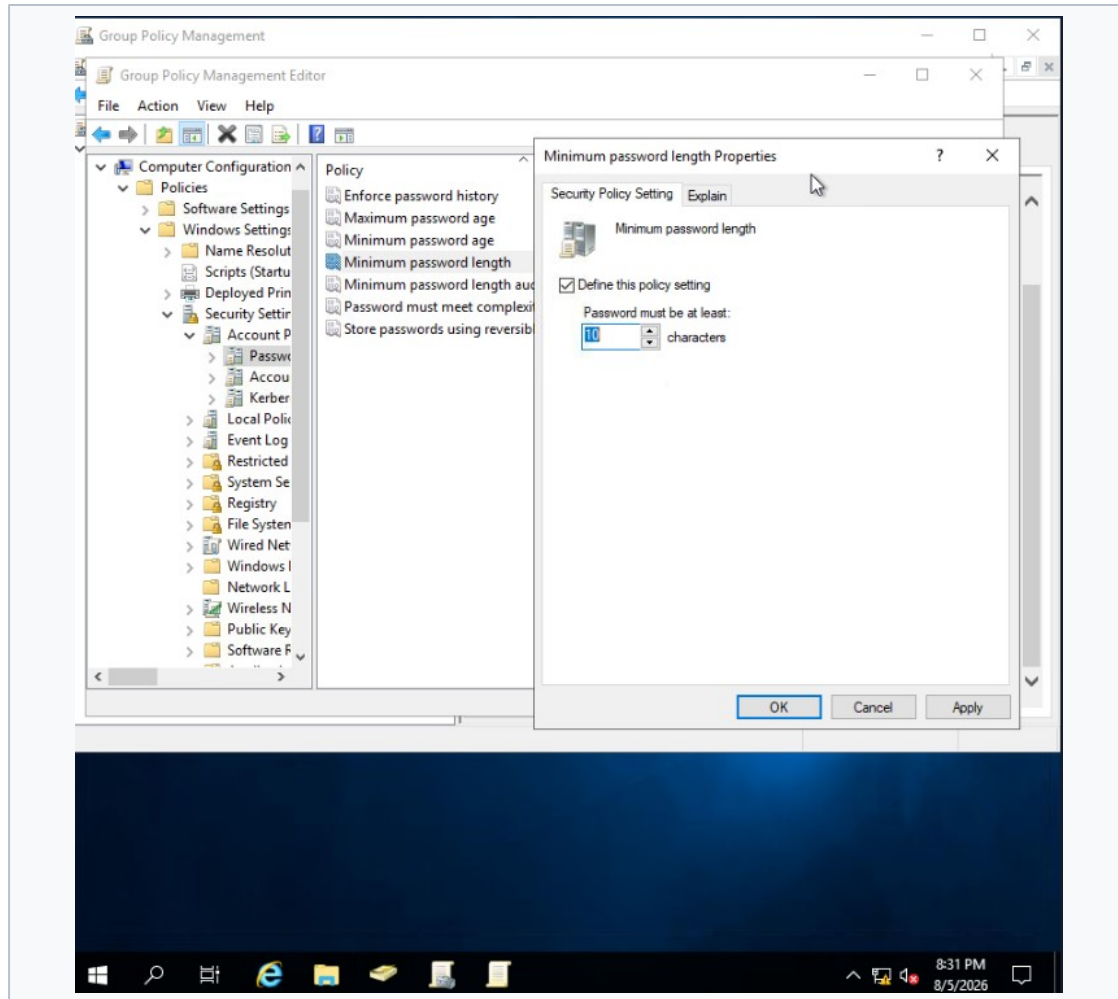
Commands used

```
Set-ADAccountPassword sophie -Reset -NewPassword  
(Read-Host -AsSecureString -Prompt 'New Password') -Verbose  
  
Set-ADUser -Identity sophie -ChangePasswordAtLogon $true -Verbose
```

Support skill demonstrated

Read the error, corrected the input, retried safely, and verified verbose success output.

I Strengthened the Domain Password Baseline to 10 Characters



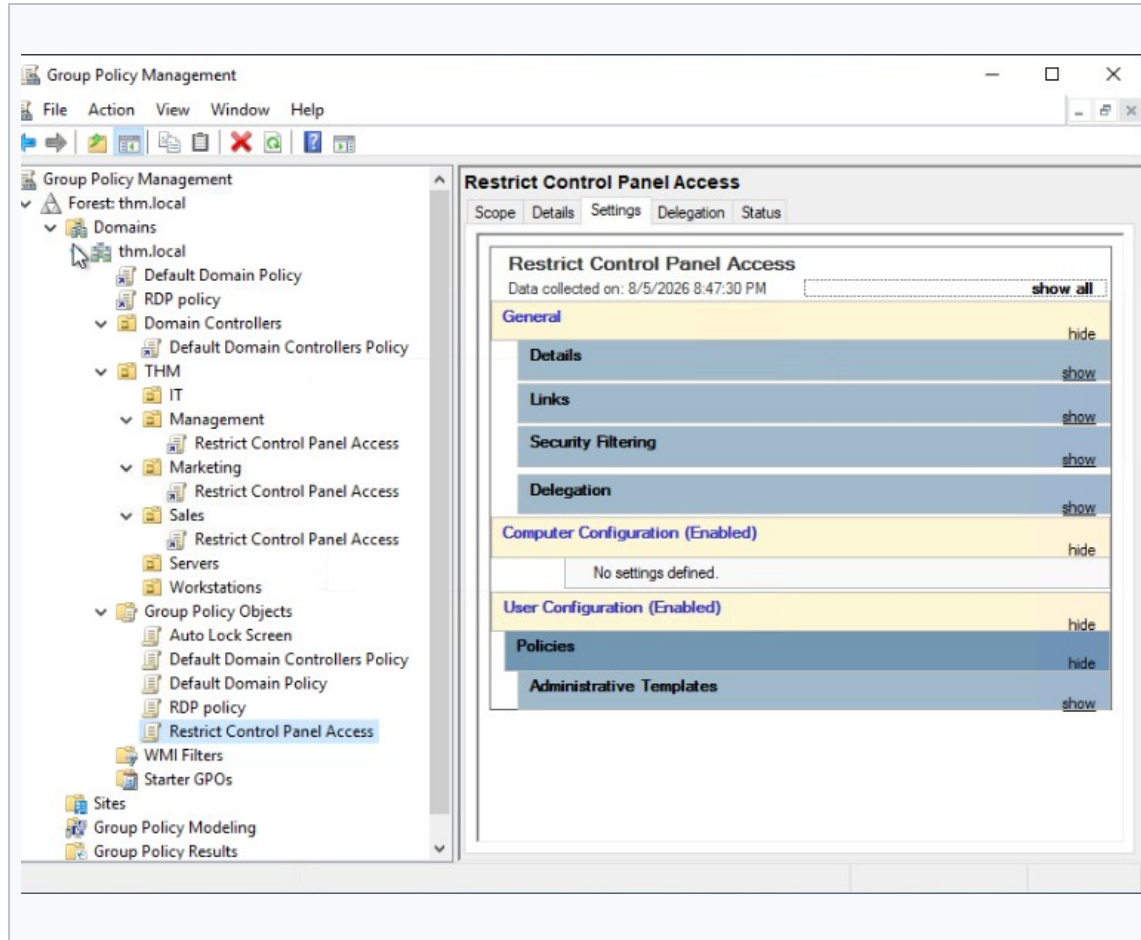
Configuration

- Opened Group Policy Management and edited the Default Domain Policy.
- Navigated to Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy.
- Changed Minimum password length from 7 to 10 characters.
- Used the Explain tab to review the policy's behavior and requirements.

Scope awareness

Domain password policy is configured at the domain level, so this setting establishes the minimum for domain accounts.

I Linked Workstation Controls at the Scope where They were Needed



Restrict Control Panel Access

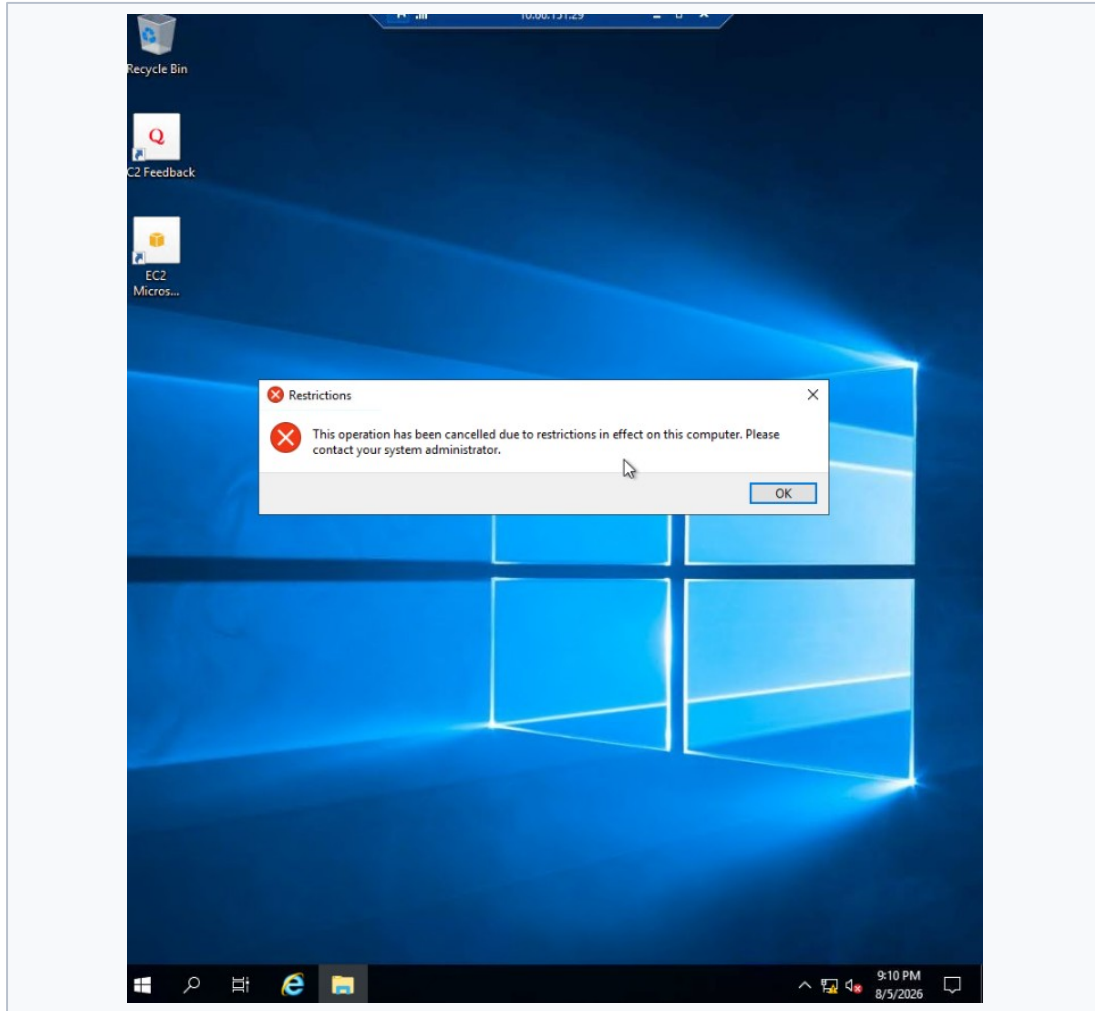
- Created a dedicated GPO and enabled Prohibit access to Control Panel and PC settings under User Configuration.
- Linked the GPO to the Management, Marketing, and Sales OUs.

Auto Lock Screen

- Created a second GPO with a five-minute machine inactivity limit.
- Linked it at the root domain so the workstation-lock setting could apply domain-wide.

Technical wording: dragging a GPO to an OU creates a link; it does not merge the GPO with the OU.

RDP Testing Confirmed the Control Panel Restriction



Test method

- Connected to the Windows host with Remote Desktop as Mark, a user in the Marketing OU.
- Attempted to open Control Panel from the user session.
- Observed the system message that the operation was canceled because of restrictions.
- Confirmed the user-scoped GPO was applied to the intended department.

Result

The control moved from configuration to verified behavior - the key difference between making a change and proving it worked.

The Project Followed an End-to-End Administration Workflow

01 ORGANIZE



Build the directory

Created and aligned OUs and user accounts to the approved department structure.

02 CONTROL



Apply least privilege

Delegated password support and configured domain and workstation policies at the correct scope.

03 VERIFY



Test the outcome

Used PowerShell verbose output and an RDP user session to confirm the intended behavior.

What I learned from the Lab

Directory design

How OUs provide an administrative and Group Policy boundary for organizing users by business role.

Least privilege

How the Delegation of Control Wizard can grant a narrowly scoped task without unnecessary administrative access.

PowerShell support

How to reset an AD account securely, interpret policy errors, and require a password change at next sign-in.

Policy lifecycle

How to configure, link, scope, and validate GPOs instead of assuming a saved setting is working.

Summary

Active Directory Administration Lab | TryHackMe VM

- Built and administered a Windows Server Active Directory lab, organizing users into role-based OUs and removing obsolete directory objects.
- Delegated password-reset permissions at the Sales OU level and used PowerShell over RDP to reset an account and require a password change at next sign-in.
- Configured and linked GPOs for a 10-character minimum password length, department-level Control Panel restrictions, and automatic workstation lock after five minutes; validated policy enforcement through user testing.

Skills: Active Directory Domain Services, ADUC, Group Policy Management, PowerShell, RDP, OU administration, delegated permissions, troubleshooting